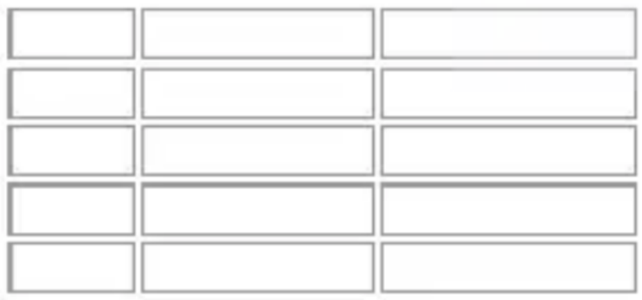
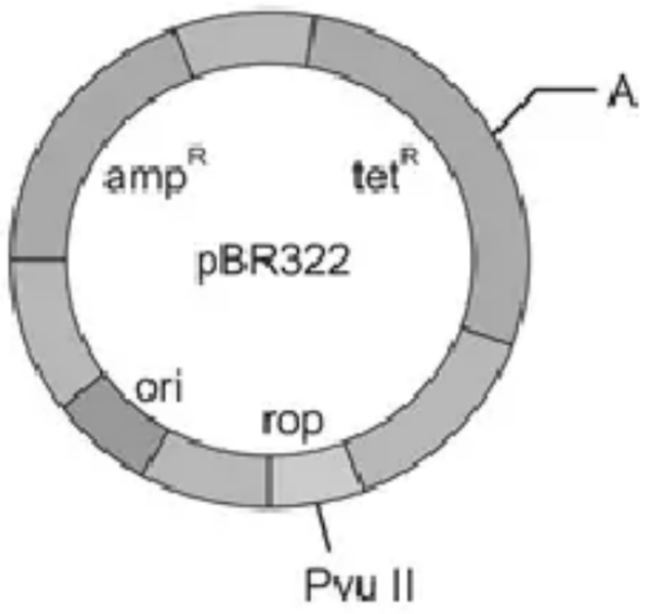
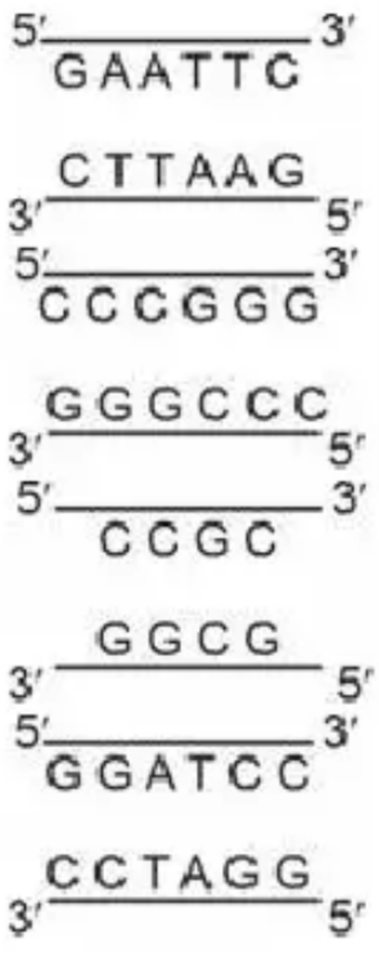
A

B




A
amp^R
tet^R
pBR322
ori
rop
Pvu II


5′ GAATTC 3′
CTTAAG
3′ 5′
5′ CCCGGG 3′
GGGCCC
3′ 5′
5′ CCGC 3′
GGCG
3′ 5′
5′ GGATCC 3′
CCTAGG
3′ 5′

168. Bacteria cell wall degrading enzyme is
(1) Cellulase
(2) Lysozyme
(3) Methylase
(4) Chitinase

169. The novel strategy used by Eli Lilly in 1983 for production of humulin by genetic engineering was
(1) Production of Pro-insulin by *E.coli* and maturation in bioreactors
(2) Production of A and B chains separately and then combined
(3) Production of chains A and B together in the same host cell and then extracted
(4) Production of mature insulin directly by *E.coli* and requiring no further combining of chains

170. Genetic modification in plants have been useful in all of the following ways, **except**
(1) Increased efficiency of mineral usage
(2) Reduced post-harvest losses
(3) Make crops more tolerant to abiotic stresses
(4) Increased reliance on chemical pesticides

171. The applications of biotechnology include all, **except**
(1) Agro-chemical based farming
(2) Production of pest resistant plants
(3) Bioremediation
(4) Therapeutics

172. Transgenic animals are used in toxicity testing of a drug for safety of human beings. In such cases, alien gene is introduced in transgenic animals to make them
(1) More sensitive to toxic substances than non-transgenic animals
(2) Resistant to toxic substances for better tolerance
(3) Less sensitive to toxic substances than normal human beings
(4) Non-reactive to toxic substances for safer use in human beings

173. The method of RNAi involves silencing of a specific $\underline{A}$ due to formation of $\underline{B}$.
Choose the option that fills the blanks **correctly.**
(1) **A** - mRNA, **B** - dsRNA
(2) **A** - ssDNA, **B** - dsDNA
(3) **A** - dsDNA, **B** - ssDNA
(4) **A** - dsRNA, **B** - mRNA

174. There are an estimated _____ varieties of rice in India out of which _____ are documented varieties of Basmati. Choose the option which correctly fills the blanks respectively.
(1) 5,00,000; 30
(2) 1,00,000; 50
(3) 2,00,000; 35
(4) 2,00,000; 27

175. Choose the **correct** statement w.r.t probe.
(1) It is always a single stranded RNA, tagged with a radioactive molecule.
(2) It is allowed to hybridise to its complementary DNA in a clone of cells followed by detection using autoradiography.
(3) It usually have the complementarity with the mutated gene.
(4) It is the single stranded DNA or RNA, not tagged with a radioactive molecule.

176. The first transgenic cow was made to produce human protein-enriched milk in the same year when
(1) The first clinical gene therapy was given
(2) An American company got patent rights on Basmati rice through the US Patent and Trademark Office
(3) An American company prepared Humulin
(4) A four-year old girl got a treatment for adenosine deaminase (ADA) deficiency

177. Given below are some reasons for the production of transgenic animals.
(a) For study of normal physiology and development
(b) Chemical safety and testing
(c) For production of useful biological products
(d) Study of diseases
(e) To test the vaccine safety
How many of the above given reasons are true ?
(1) One
(2) Five
(3) Three
(4) Two

178. Enzyme-Linked Immunosorbent Assay (ELISA) is based on the principle of
(1) Antigen - antibody interaction
(2) Antigen - antigen interaction
(3) Antibody - antibody interaction
(4) Antibody - lipid interaction

179. In case of *Bam*HI, '*am*' represents the name of
(1) Genus
(2) Species
(3) Strain
(4) Phylum

180. Read the following statements.
Statement A : The normal *E.coli* does not multiply in the ampicillin containing medium.
Statement B : Transgenic models exist for many human diseases such as cancer, cystic fibrosis, rheumatoid arthritis and Alzheimer's.
Choose the **correct** option.
(1) Both statements are correct
(2) Both statements are incorrect
(3) Only statement A is correct
(4) Only statement B is correct

181. In the case of SCID, which technique is considered to be completely curative?
(1) Oral intake of ADA tablets
(2) Bone marrow transplantation
(3) Enzyme replacement therapy
(4) Gene therapy in early embryonic stage

182. Which one of the following proteins obtained from transgenic animals is being used to treat emphysema?
(1) Insulin
(2) α-1-antitrypsin
(3) Interferon
(4) α- lactalbumin

183. Abnormal gene is replaced by normal gene through
(1) Cloning
(2) Radiation
(3) Gene therapy
(4) Surgery

184. In India, at present how many recombinant therapeutic are being marketed?
(1) 12
(2) 30
(3) 27
(4) 40

185. The full form of EFB is
(1) Eastern Federation of Bioprocess Engineering
(2) Eastern Federation of Bioprocess Biotechnology
(3) European Federation of Biotechnology
(4) European Federation of Bioprocess Engineering

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SECTION-B

186. How many interchain disulphide bond(s) are present in proinsulin?
(1) One
(2) Two
(3) Three
(4) Zero

187. Which of the following statements is not **incorrect** w.r.t Bt cotton?
(1) Some strains of *Bacillus thuringiensis* produce proteins that kill certain nematodes
(2) Bt toxin protein exists as active protoxins but once an insect ingest the active toxin, it is converted into an inactive form
(3) The proteins encoded by the genes *cry IAc* and *cry II Ab* control the corn borer and that *cry IAb* controls cotton bollworms
(4) Bt toxin gene has been cloned from the bacteria and been expressed in plants to provide resistance to insects without the need for insecticides; in effect created a bio-pesticide

188. In biotechnology, antibiotic resistance genes are used
(1) As selectable markers
(2) As vectors
(3) To maintain sterile condition in culture medium
(4) As enzyme resistant markers

189. In gel electrophoresis,
(1) Smaller fragments travel slower than larger fragments
(2) Larger DNA fragments stay closer to cathode as compared to smaller fragments
(3) Ethidium bromide is used to visualise DNA bands in visible light
(4) DNA is extracted from gel by spooling

190. Downstream processing in biotechnology includes
(1) Introduction of recombinant DNA into host cells
(2) Large scale culture of recombinant bacteria in bioreactors
(3) Isolation of DNA from cell
(4) Separation and purification of recombinant protein

191. Stirrer in stirred-tank bioreactor has been designed for
(1) Providing a temperature control system
(2) Ensuring anaerobic conditions in culture vessel
(3) Addition of preservatives to the product
(4) Facilitating the even mixing and availability of oxygen throughout the process

192. If a circular DNA possesses 6 restriction sites for *Bam*HI then how many fragments will be produced upon their digestion with same restriction enzyme?
(1) 6
(2) 7
(3) 8
(4) 9

193. Read the features w.r.t. cloning vectors.
(A) *ori* site
(B) Selectable marker
(C) Cloning site
(D) Recognition site
Select the set which is correct for the features of cloning vectors.
(1) (A) and (B) only
(2) (A), (B) and (C) only
(3) (B) and (C) only
(4) (A), (B), (C) and (D)

194. Agarose is a natural polymer extracted from
(1) Fungus
(2) Bacteria
(3) Sea weeds
(4) Protozoan

195. In the following question, a statement of assertion (A) is followed by a statement of reason (R).
A : Restriction enzymes link foreign DNA to a plasmid in order to produce rDNA.
R : Restriction enzymes are called molecular glue.
In the light of above statements, select the correct option.
(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion
(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion
(3) Assertion is true statement but Reason is false
(4) Both Assertion and Reason are false statements

196. In RNAi, host-generated dsRNA triggers protection against nematode infestation as it prevents
(1) Formation of sense-RNA
(2) Translation of a specific mRNA
(3) Formation of antisense-RNA
(4) Lipid synthesis in host plant

197. Bt toxins are
(1) Insect-group specific
(2) Human-group specific
(3) Crop-group specific
(4) Fertiliser-group specific

198. Illegal and unlawful utilisation of biomaterials without payment to people concerned is called
(1) Bio-war
(2) Biopatent
(3) Biopiracy
(4) Biotechnology

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199. Select the **correct** statement.

(1) RNAi is a post transcriptional gene silencing method

(2) Golden rice is a drought tolerant variety developed using *Agrobacterium* vector

(3) Human insulin can be produced by bacteria as genetic code is never universal and is species specific

(4) In upstream processing, addition of preservatives in final product plays an essential role

200. Which of the following is **mismatched**?

(1)	Rosie	–	Human protein - enriched milk (2.4 grams per litre)
(2)	Serum analysis	–	Conventional method of disease diagnosis
(3)	PCR	–	Used to check the progression of restriction enzyme digestion
(4)	ELISA	–	Early diagnosis of diseases

(1) (1)

(2) (2)

(3) (3)

(4) (4)

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